

tool. It checks a single Web page against the accessibility standards. Being a Web based tool, it can be accessed via the URL <https://achecker.ca>.

To check a site for accessibility, enter this URL in the 'Address' text box and click on 'Check it'. You can either upload an HTML file or directly enter the HTML markup in the window provided for that. You will get a result indicating the accessibility violations on that page. The results are classified into the following categories:

- Known problems
- Likely problems
- Potential problems

A sample result page is shown in Figure 4.

The reports can be extracted to various formats such as CSV, HTML or PDF.

AChecker Web service: If you want to access the site through code, AChecker provides a Web service interface as well. The following parameters need to be used.

- *uri*: This is the URL of the Web page to be validated.
- *id*: This is the Web service ID that can be obtained by registering with AChecker. You can retrieve your ID from the profile page.
- *Guide*: This includes the set of rules that need to be validated (WCAG, Stanca, etc).
- *output*: This indicates the output parameters.
- *offset*: You can provide this if you specify the line offset to start the validation.

A sample request is shown below:

```
https://achecker.ca/checkacc.php?uri=https://opensourceforu.com/& id=888ca9e3f856baa0120755ecd8ffae6be3142029&output=html&guide=WCAG2-AA&offset=10
```

A portion of the output template is shown below:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<resultset>
  <summary>
    <status>FAIL</status>
    <sessionID>40-character-long
string</sessionID>
    <NumOfErrors>number</NumOfErrors>
    <NumOfLikelyProblems>number</
NumOfLikelyProblems>
    <NumOfPotentialProblems>number</
NumOfPotentialProblems>

    <guidelines>
      <guideline>string</guideline>
      ...
    </guidelines>
  </summary>
```

Tips to build accessible pages

To introduce bare minimum accessibility into your pages, do the following:

- While selecting a content management system (CMS), choose the one with the best accessibility features.
- Proper usage of headings ensures that content is organised in a proper structure.
- Provide text alternatives to all non-text elements; for example, all images must have *alt-text*, a textual description of the image.
- Provide proper descriptive names for all the links in your page. Repeated usage of terms such as 'Click Here' should be avoided.
- Provide features to adjust text size.
- Ensure proper contrast. Colours should be selected with care.
- Make sure the form elements

in your page are accessible by properly labelling them. Proper tab navigation should be provided.

- While using captchas, choose one with accessibility features. Many a time, captchas are barriers to accessibility.
- Avoid using tables for content layout, as they are difficult for screen reader users to understand. Use them only for providing tabular data.
- Ensure proper keyboard navigation, i.e., users who can't use a mouse should also be able to navigate through your site without difficulties.
- Use ARIA (Accessible Rich Internet Applications) roles and landmarks.
- If you have video content in your pages, ensure that they don't auto-play. Provide closed captions for all the videos. Give users the option to operate the media player via the keyboard.

Web accessibility is now a field of active research, and falls under the Human Computer Interaction (HCI) domain. The field of accessibility computing, which deals with making the digital ecosystems accessible, is expanding rapidly. What has been presented in this article is an introduction to Web accessibility. If you are interested in the accessibility domain, there are plenty of resources available to explore further. Together, let's build an inclusive Web. **END** 🐧

References

- [1] <https://www.w3.org/WAI/>
- [2] <https://www.w3.org/WAI/ER/tools/?q=wcag-20-w3c-web-content-accessibility-guidelines-20>
- [3] <https://achecker.ca/checker/index.php>

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